

1. Single Matrix Implementation in Multiclass Classification:

the idea is to organize the weights for all neurons in a layer into a single matrix, where each row in the matrix represents a set of weights for a neuron.

take the single matrix to implement a matrix multiplication with "another matrix" and compute other necessary matrix multiplications.

this is, we get similar result compared to non-matrix implementation,

but it runs faster because of more efficient implementation using matrix multiplication instead of for-loop multiplication.

2. Mini Batch Implementation in Multiclass Classification:

we can divide all input/output examples in dataset into multiple sets,

thereby, we get batch size of input/output examples per set.

we organize each set into a matrix, where each row in the matrix represents an input/output example.

finally, we use the matrix above as "another matrix" in Single Matrix Implementation and run Single Matrix Implementation.